# Analysis of an example from Ascher & Petzold

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This note discusses an example that appears on p. 264 of *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations* involving the linear DAE system:

$$\begin{pmatrix} 0 & 0 \\ 1 & \eta t \end{pmatrix} \mathbf{y}' + \begin{pmatrix} 1 & \eta t \\ 0 & 1+\eta \end{pmatrix} \mathbf{y} = \begin{pmatrix} q(t) \\ 0 \end{pmatrix}. \tag{1}$$

The exact solution is $y_1(t) = q(t) + \eta t q'(t)$, $y_2(t) = -q'(t)$.

If we apply backward Euler to (1), we get

$$\begin{pmatrix} 0 & 0 \\ 1 & \eta t_n \end{pmatrix} \frac{\mathbf{y}_n - \mathbf{y}_{n-1}}{h} + \begin{pmatrix} 1 & \eta t_n \\ 0 & 1+\eta \end{pmatrix} \mathbf{y}_n = \begin{pmatrix} q(t_n) \\ 0 \end{pmatrix}, \tag{2}$$

as stated in the text. We then define

$$\begin{pmatrix} u_n \\ v_n \end{pmatrix} = \begin{pmatrix} 1 & \eta t_n \\ 0 & 1 \end{pmatrix} \mathbf{y}_n. \tag{3}$$

Then the first equation of (2) becomes simply

$$u_n = q(t_n), \tag{4}$$

which is consistent with the exact solution. It is also claimed that we have

$$(1+\eta)v_n = \frac{q(t_n) - q(t_{n-1})}{h}.$$

In fact, the second equation of (2) reads

$$\frac{y_{1,n} + \eta t_n y_{2,n}}{h} - \frac{y_{1,n-1} + \eta t_n y_{2,n-1}}{h} + (1+\eta) y_{2,n} = 0. \tag{5}$$

Using (3), the first term above is simply $u_n/h$. If we replaced the second term by $u_{n-1}/h$, we would get the claimed result (after using also $u_n = q(t_n)$ and $v_n = y_{2,n}$).

However, the second term is not equal to $u_{n-1}/h$, because it involves $t_n$ rather than the needed $t_{n-1}$. In fact, the second term is equal to (since $v_n = y_{2,n}$)

$$\frac{u_{n-1}}{h} + \frac{\eta v_{n-1}(t_n - t_{n-1})}{h} = \frac{u_{n-1}}{h} + \eta v_{n-1}.$$

Thus (5) simplifies to

$$\frac{u_n - u_{n-1}}{h} - \eta v_{n-1} + (1+\eta) v_n = 0, \tag{6}$$

which, using (4), gives

$$v_n + \eta(v_n - v_{n-1}) = -\frac{q(t_n) - q(t_{n-1})}{h}. \tag{7}$$

For small $h$ this is

$$v_n = -q'(t_n) + O(h), \tag{8}$$

a consistent discretization of the exact solution $y_2(t) = -q'(t)$.

The purpose of this example is to convince the reader that directly discretizing DAEs of index greater than 1 is “not recommended”. I’m not an expert, but I guess this advice is still correct even if the example is not. In that case, it would be helpful to have a correctly-worked example that shows what can go wrong.